# Run record: nine-arm held-out fibers

**Date:** 2026-10-16. **Host:** Darwin arm64 shared L0. **Revision:** `codex/isogeny-portable-trend-20261015`. The frozen V4 spec records exact source, binary, meter, orchestration, public and checker-only input SHA-256 values.

**Hypothesis:** inverse-Vélu and full-image constructions select the published destination base and have the same relation support as the χ, direct-pair and source/quotient S3 arms. Their cold cost may differ. See `HELDOUT_NINE_ARM_PROTOCOL_V4.md`.

**Commands:** `bash RUN_HELDOUT_NINE_ARM_V4.sh` created 108 separate native child and meter records. `different_j_chi_heldout_check` replayed both curves, 996 relation witnesses, row ranks, BSGS and rho. The native summary asserted identical six-arm trial and relation support. All checks passed.

**Preserved correction:** an initial compilation hit `json!`'s recursion limit while building the new inverse child output. Splitting the nested value fixed compilation before the V4 run.

**Limits:** 300-second hard wall cap per child and 512 MiB observed Darwin RSS ceiling. No child failed or timed out. The host has no isolation or thermal receipt. Complete portable S and admitted wall time remain open.

See `HELDOUT_RESULTS_V4.md`, `HELDOUT_NINE_ARM_V4.summary.json`, `HELDOUT_NINE_ARM_V4.check.json`, the raw directory and hash manifest.